02/18/2022

Thursday, February 17, 2022 9:55 PM

M168_Notes_6.1



# Chapter 6: SETS AND COUNTING

## 6.1 Sets and Set Operations

A ***set*** is a collection of objects.

The objects in a set are the ***elements*** or ***members*** of the set.

→ Always enclose the elements of a set in curly brackets.

A set with the numbers $\{-1,1,0\}$ would be written as

where is read "is an element of"

Define $S = \{-1, 1, 0\}$ then $\boxed{1 \in S}$

"1 belongs to S"

$\phi = \{\ \}$

$\{\phi\} = \{\{\ \}\}$

More notation:

- $0$ is the symbol for the real number zero
- $\{0\}$ is a set with one element, the real number zero
- $\varnothing$ is a set with zero elements, the empty set. Alternative is $\{\}$. → $\phi$ - phi
- $\{\varnothing\}$ is a set with one element, the symbol for the empty set.

Two sets are ***equal*** (=) if they contain exactly the same elements (order doesn't matter).

$S = \{-1, 1, 0\}$

$R = \{0, 1, -1\}$

$R = S$

They are ***not equal*** (≠) if they don't contain the same elements.

$R_2 = \{+1, 0, 2\}$

$R_2 \neq S$

$R_3 = \{\{-1, 1, 0\}\}$

Then $R_3 \neq S$.

***Set builder notation:*** Describe the set in terms of its properties,

$A = \{x \mid$ Condidacy statement$\}$. e.g. $\Rightarrow A = \{x \mid 0 \leq x \leq 13,\ x$ is an integer$\}$.

***Roster notation:*** List the elements of the set.

Non example: $A = \{x \mid x$ is a balled guy$\}$ ✗

$A = \{x \mid x$ is false$\}$ ✗

$A = \{0, 1, 2, \ldots, 13\} = \{13, 12, 11, \ldots, 0\}$

***Subset:*** Set $B$ is a subset of set $A$ (written $B \subseteq A$) if every element in $B$ is in $A$.

$x \in B, \Rightarrow x \in A.$, E.g. $A = \{0, 1, \ldots, 13\}$

$\rightarrow B = \{0, 2, 4, 6, \ldots, 12\}$ Then $B \subseteq A$.

$\rightarrow B = \{x \mid 0 \leq x \leq 13$ and $x$ is divisible by $2\}$.

12

***Proper Subset:*** Set $B$ is a proper subset of set $A$ (written $B \subset A$) if $B \subseteq A$ and $A \neq B$.

A set $S$. Then $S \subseteq S$ always.

$B \subseteq A$ and $B \neq A$.

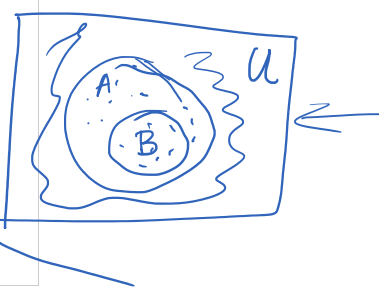
$B$ is a pooper subset of $A$, will denote $B \subset A$.

***Universal set:*** The set from which all the member of other sets will be drawn. Called $U$.

$U = \{x \mid x$ is a non-neg. integers less than $1700\}$.

***Venn Diagram notation:***

- A rectangle represents the universal set
- Circles are sets in the universal set.




*Example*

Show the relationship between $A$ and $B$ (defined above) in a Venn diagram.

Given a set $A$ and a universal set $U$, the elements that are in $U$ and are NOT in $A$ is called the ***complement*** of $A$ or $A^c$.

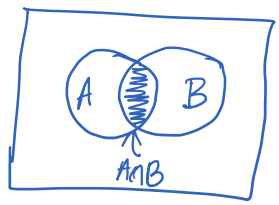
$A^c = \{x \mid x \in U \text{ and not in } A\}$, e.g. $A^c = \{x \mid x \text{ is a non-neg. integer and } 13 < x < 1700\}$.

*Example*

From the last example, $A$ is the set of even integers, what is $A^c$ in roster notation?

Those elements that belong to both $A$ ***and*** $B$ are in the ***intersection*** of $A$ and $B$, $A \cap B$.

$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$.




*Example*

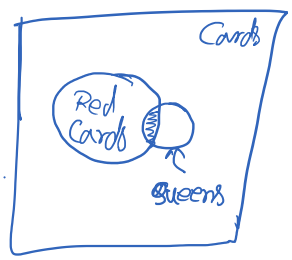
Let $U = \{x \mid x \text{ is a card in a standard deck of 52 playing cards}\}$

$R = \{x \mid x \text{ is a red card}\}$

$Q = \{x \mid x \text{ is a queen}\}$

Find $R \cap Q$ in roster notation.

$R \cap Q = \{x \mid x \text{ is a red card and } x \text{ is a queen}\}$.



If two sets have no elements in common, that is $A \cap B = \varnothing$, then the sets are ***disjoint***.

Empty.

*Example*
If $K = \{x|x \text{ is a king}\}$, find $K \cap Q$ in roster notation.

Those elements that belong to $A$ ***or*** $B$ are in the ***union***, $A \cup B$.

$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$.

$R \cup Q = \{x \mid x \text{ is a red card or } x \text{ is a queen}\}$.

Note: this is the ***inclusive or***, not the exclusive or

*Example*
Let $U = \{x|x \text{ is a card in a standard deck of 52 playing cards}\}$
$H = \{x|x \text{ is a heart card}\}$
$Q = \{x|x \text{ is a queen}\}$

$|H| = 13$ $n(H) = 13$
$|Q| = 4$ $n(Q) = 4$, $n((H \cup Q)) = 16$

Find $H \cup Q$ in roster notation.

$H \cup Q = \{AH, 2H, 3H, \ldots, 10H, KH, QH, JH, QS, QC, QD\}$

$|H \cup Q| = 16$

$H \cap Q = \{QH\}$

$n(H \cap Q) = 1$

$16 = 13 + 4 - 1$

*Example*

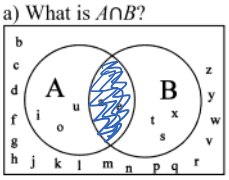
Let $U = \{x|x \text{ is a letter in the English alphabet}\} = \{a, b, c, ..., z\}$

$A = \{x|x \text{ is a vowel}\} = \{a, e, i, o, u\}$

$B = \{x|x \text{ is a letter in the word texas}\} = \{ t, e, x, a, s\}$

Find the following sets in roster notation.

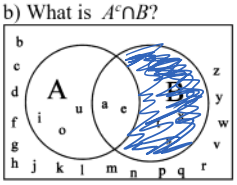
a) What is $A \cap B$?




$A^c = \{x|$ letter, which is not a vowel$\}$.

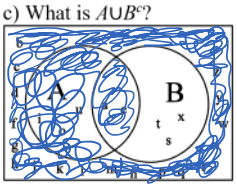
b) What is $A^c \cap B$?

$\rightarrow A^c \cap B$




c) What is $A \cup B^c$?

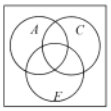
$\rightarrow B^c \cup A$



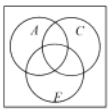
*Example*

People at a home show were surveyed to see if they planned on replacing their kitchen countertops ($C$), their kitchen floor ($F$) or their kitchen appliances ($A$). Shade the following regions on the Venn diagram and express the region in set notation.

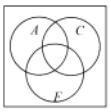
People who planned to replace their countertops and floor



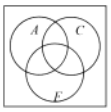

People who planned to remodel all three features

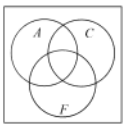



People who planned to replace their appliances or countertops




People who planned to replace exactly one of these features




People who planned to replace their kitchen appliances or floor, but not their countertops.